

Routing And Switching Essentials II





COURSE STRUCTURE

Day 1: Ethernet Switching, VLANs, Routing, OSPF & EIGRP

Day 2: STP, EtherChannel, HSRP, DHCP, NAT & Data Center Architecture

Day 3: Networking in the Cloud Era — AWS VPC, EC2 & Security

Day 4: Cloud Network Services & Hybrid Connectivity

Day 5: High Availability, Troubleshooting & Network Management Platforms

AGENDA FOR DAY 2

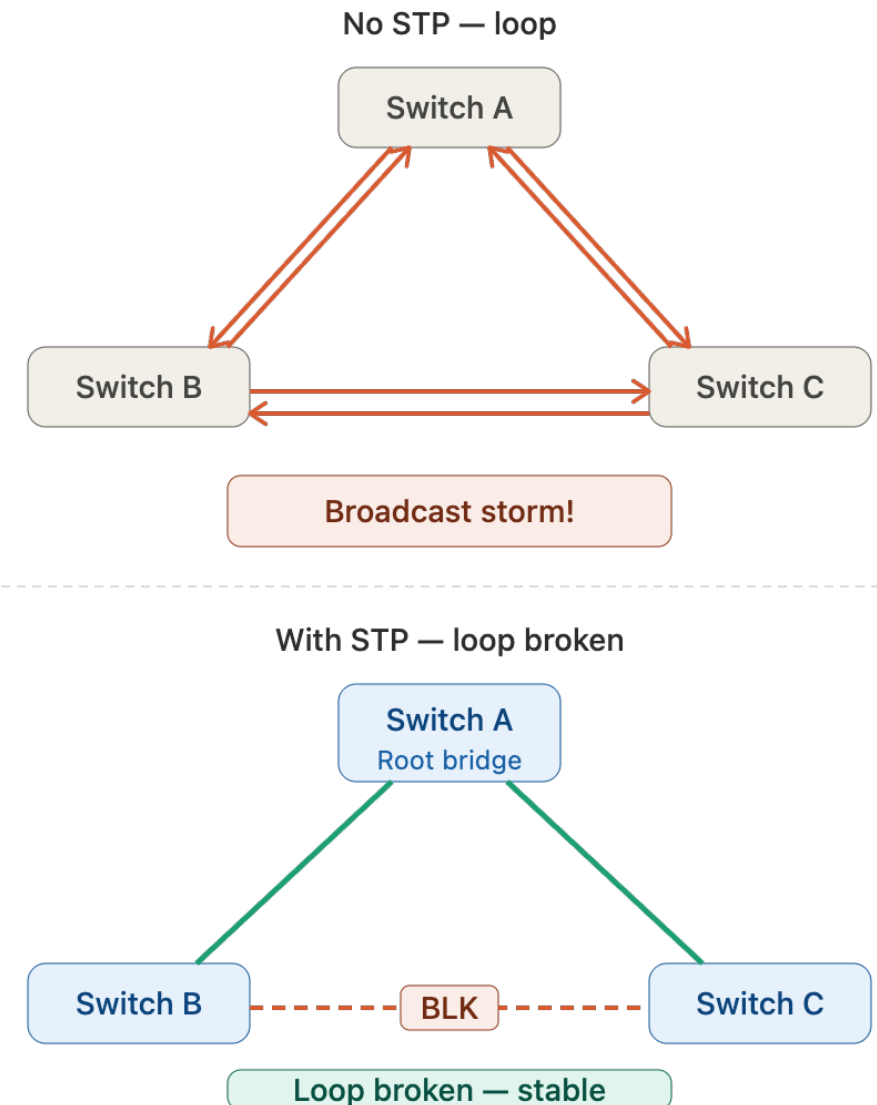
1. Spanning Tree Protocol — Loop Prevention, Root Election, Port Roles & States
2. Rapid STP, PortFast & BPDU Guard
3. EtherChannel / LACP — Link Aggregation Configuration & Verification
4. First-Hop Redundancy — HSRP & VRRP
5. DHCP — Scopes, Relay & Troubleshooting
6. NAT — Static, Dynamic & PAT
7. Data Center Architecture — Three-Tier & Spine-Leaf



SPANNING TREE PROTOCOL

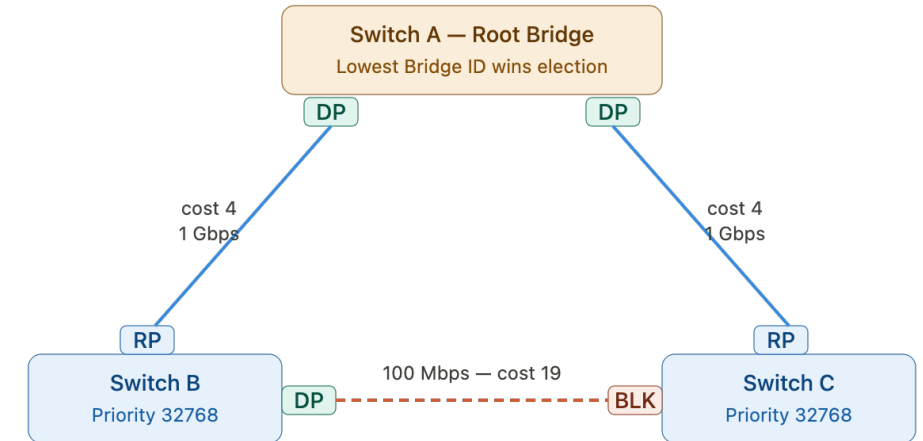
WHY STP EXISTS — THE LOOP PROBLEM

- Redundant L2 links are essential for availability — but without STP they create catastrophic loops
- Broadcast storm: switch A floods → B receives and floods back → A floods again → infinite loop — network destroyed
- STP (IEEE 802.1D) prevents loops by logically blocking redundant ports — physically connected but not forwarding
- A blocked port continuously monitors BPDUs and can unblock in ~30 seconds if the primary path fails
- Without STP: connecting a single redundant cable to a live network can bring down the entire L2 domain



STP ROOT BRIDGE ELECTION & PORT ROLES

- Root Bridge: switch with the lowest Bridge ID (Priority + MAC) — the central reference point for the spanning tree
- Bridge Priority default: 32768 — lower is better; set intentionally with spanning-tree vlan 10 priority 4096
- All ports on the Root Bridge are Designated Ports (Forwarding) — traffic always flows TO the Root
- Root Port (RP): one per non-root switch — the port with the lowest cumulative cost path to the Root Bridge
- Designated Port (DP): one per network segment — forwards traffic toward the Root on that segment
- Alternate (Non-Designated) Port: all remaining ports — placed in Blocking to prevent loops



Port roles

Root bridge All ports = DP Always forwarding	Root port (RP) One per non-root Lowest cost to root	Designated (DP) One per segment Forwards to root	Blocking (BLK) All remaining ports Prevents loops
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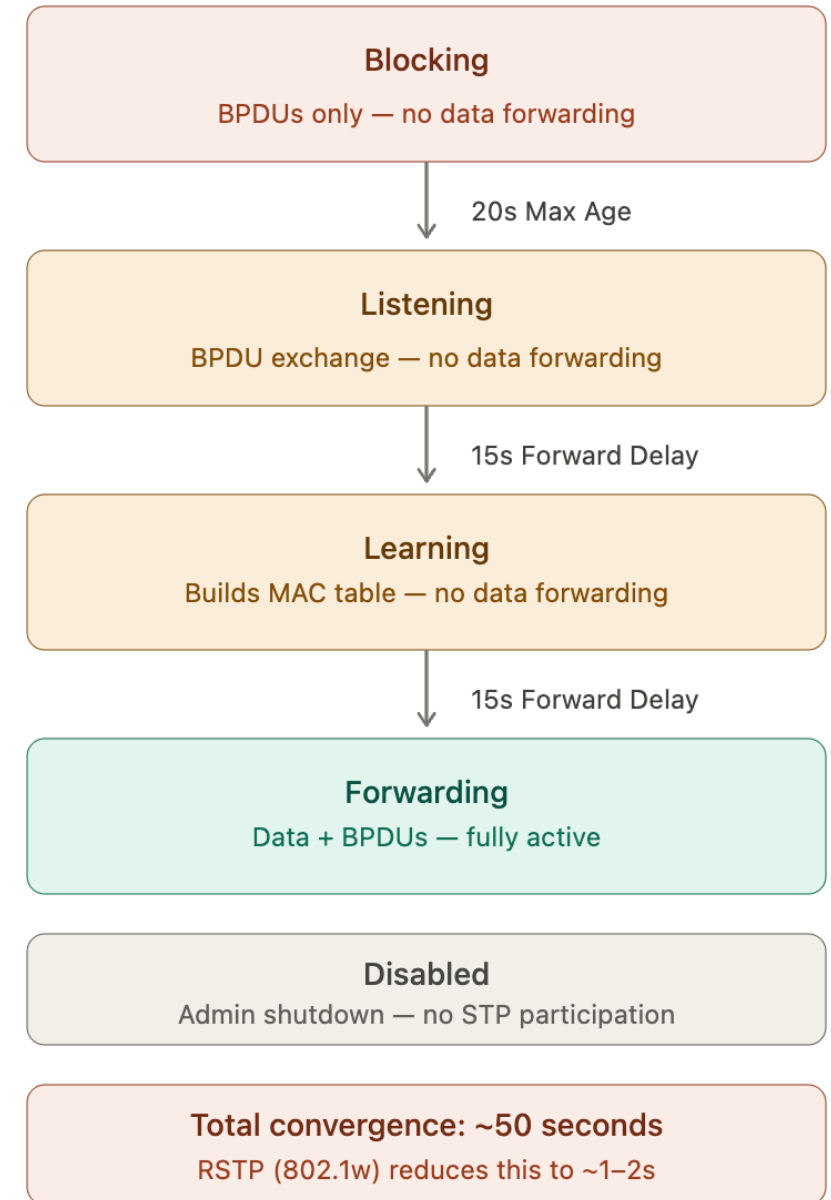
Port cost — lower = preferred path

10 Gbps = 2	1 Gbps = 4	100 Mbps = 19	10 Mbps = 100
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Election: lowest priority wins → tie-break on lowest MAC
Default priority 32768 | spanning-tree vlan 10 priority 4096

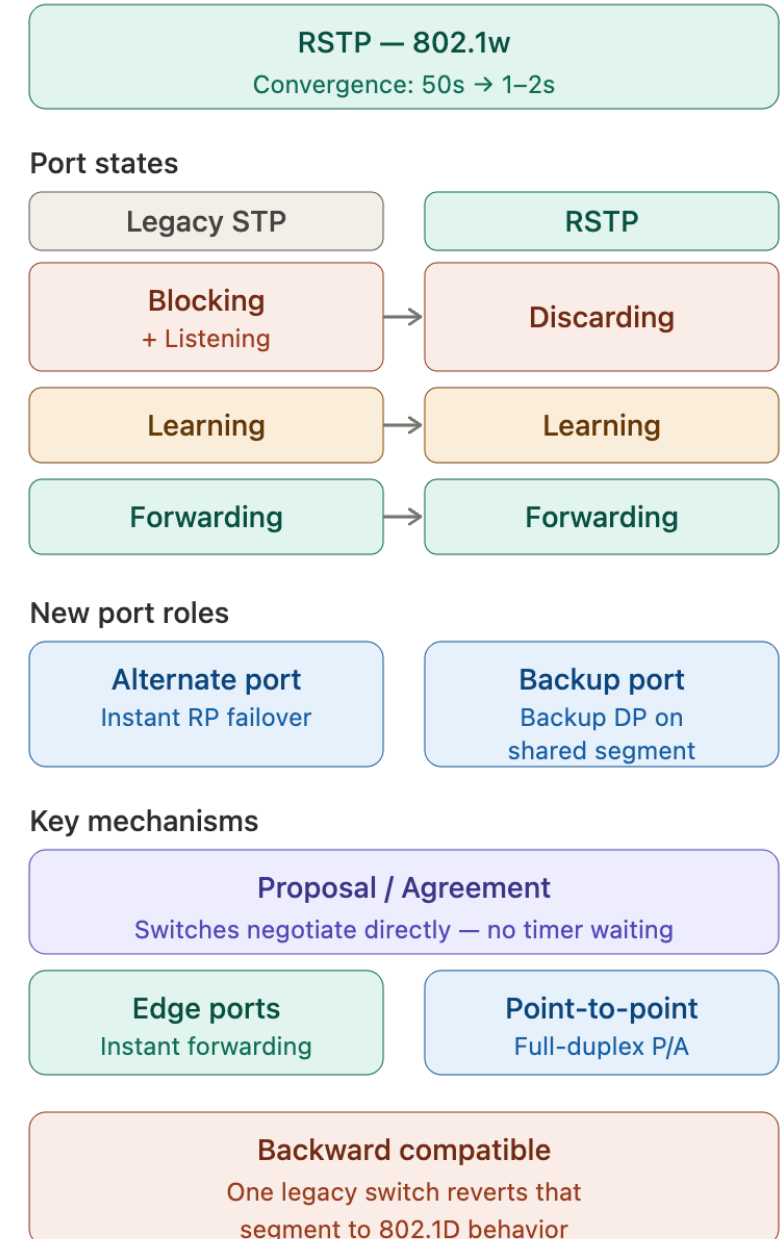
STP PORT STATES & TIMERS

- Blocking: receives BPDUs only — does NOT forward data frames; prevents loops
- Listening: participates in BPDU exchange to determine role — no data forwarding (15 seconds)
- Learning: builds MAC table from incoming frames — no data forwarding yet (15 seconds)
- Forwarding: fully active — forwards data frames and BPDUs
- Disabled: administratively shutdown — no STP participation
- Total legacy convergence: up to 50 seconds (20s Max Age + 15s Listening + 15s Learning)
- Timers: Hello=2s | Max Age=20s | Forward Delay=15s per state — all configurable on Root Bridge



RAPID STP (RSTP / 802.1W)

- RSTP reduces convergence from up to 50 seconds down to 1–2 seconds
- Simplified port states: Discarding (Blocking+Listening combined), Learning, Forwarding
- New port roles: Alternate (instant failover backup for Root Port), Backup (backup for DP on shared segment)
- Proposal/Agreement: switches negotiate topology directly instead of passively waiting for timers
- Edge ports (PortFast equivalent): immediately enter Forwarding — no delay for end-device ports
- Point-to-point links (full-duplex): converge immediately via P/A; shared links fall back to STP-style timers

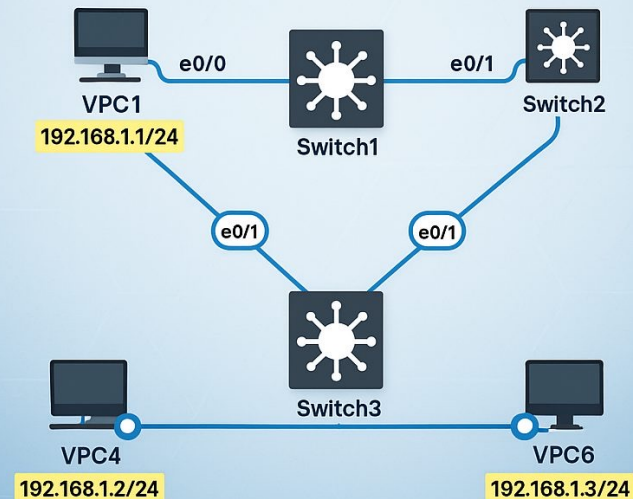


PORTFAST & BPDU GUARD

- PortFast: skips Listening and Learning states — access port immediately enters Forwarding
- Use ONLY on ports connected to end devices (PCs, printers, servers) — never on switch-to-switch links
- Per-interface: spanning-tree portfast | Globally on all access ports: spanning-tree portfast default
- BPDU Guard: if ANY BPDU is received on a PortFast port → immediately err-disable the port
- Per-interface: spanning-tree bpduguard enable | Global: spanning-tree portfast bpduguard default
- show spanning-tree inconsistentports — lists all ports currently err-disabled by BPDU Guard

Spanning Tree PortFast

Bypasses Listening & Learning States on Access Ports



POP QUIZ:

What is the primary purpose of Spanning Tree Protocol?

- A. Accelerate MAC address learning across switches
- B. Prevent Layer 2 switching loops in redundant topologies
- C. Synchronize VLAN databases across the network
- D. Load-balance traffic across all available uplinks simultaneously



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POP QUIZ:

PortFast should ONLY be configured on which type of port?

- A. Switch-to-switch trunk links
- B. Ports connected to end devices such as PCs and printers
- C. Uplinks to distribution switches
- D. Any port with a high-speed connection



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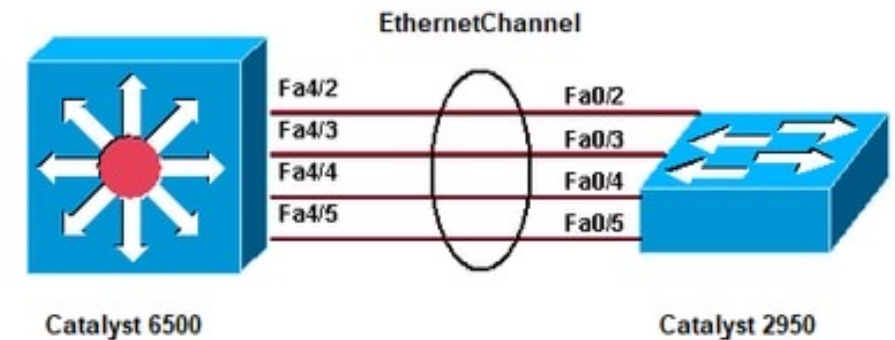
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ETHERCHANNEL / LACP

ETHERCHANNEL OVERVIEW

- EtherChannel bundles 2–8 physical interfaces into one logical Port-Channel — more bandwidth + redundancy in one link
- STP sees the entire bundle as a SINGLE logical link — no blocking within the bundle; no wasted bandwidth
- Member link failure: traffic shifts to remaining members sub-second — no STP recalculation required
- LACP (IEEE 802.3ad): open standard — works across vendors; Active mode initiates, Passive mode responds
- PAgP (Cisco proprietary): Desirable mode initiates, Auto mode responds
- Load balancing: hash based on src-mac, dst-mac, src-dst-ip, etc. — set with port-channel load-balance



ETHERCHANNEL CONFIGURATION & VERIFICATION

- Step 1: verify interfaces match — same speed, duplex, VLAN assignments, and trunk settings
- Step 2: interface range Fa0/1-2 → channel-group 1 mode active (LACP active)
- Step 3: interface port-channel 1 → switchport mode trunk → switchport trunk allowed vlan 10,20,99
- show etherchannel summary — protocol, member states: P=in bundle ✓ | I=individual (not bundled) X | s=suspended X
- show etherchannel port-channel — load-balance method and member interface details
- show interfaces port-channel 1 — logical interface statistics (bandwidth, errors, utilization)
- Common failures: mismatch in speed/duplex/VLAN on members | incompatible modes (Active↔On) | different native VLANs

POP QUIZ:

Which EtherChannel protocol is the IEEE open standard preferred in multi-vendor environments?

- A. PAgP (Cisco proprietary)
- B. LACP (IEEE 802.3ad)
- C. DTP (Dynamic Trunking Protocol)
- D. VTP (VLAN Trunking Protocol)



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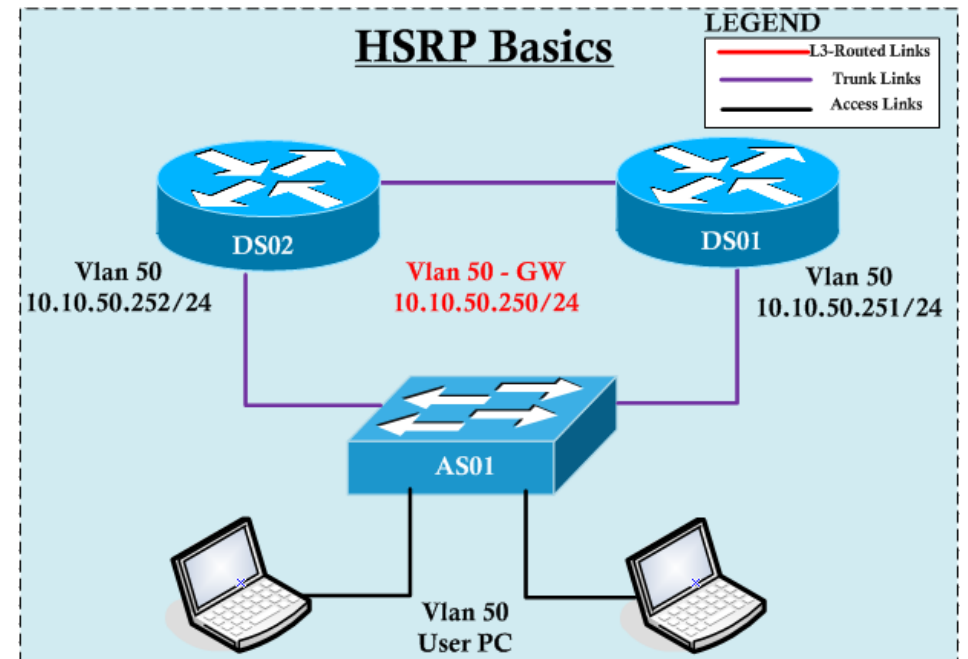
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FIRST-HOP REDUNDANCY: HSRP & VRRP

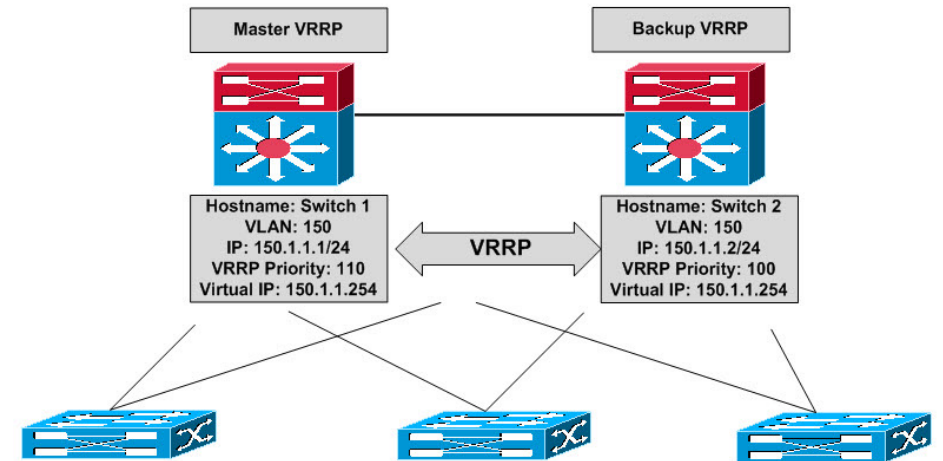
HSRP — HOT STANDBY ROUTER PROTOCOL

- Cisco proprietary — v1 and v2 available
- standby 1 ip 192.168.10.1 (virtual gateway IP)
- standby 1 priority 110 (higher = Active)
- standby 1 preempt (reclaim Active role on recovery)
- standby 1 track Gi0/0 20 (reduce priority by 20 if WAN fails)
- show standby brief — state, virtual IP, Active/Standby



VRRP — VIRTUAL ROUTER REDUNDANCY PROTOCOL

- Open standard (RFC 5798) — multi-vendor: Cisco, Juniper, Arista, HP all support VRRP
- Master router forwards; Backup routers monitor via VRRP Advertisement messages
- Virtual IP can be the same as the Master router's real interface IP — unlike HSRP
- Config: `vrrp 1 ip 192.168.10.1 → vrrp 1 priority 110`
- Preemption enabled by default in VRRP — unlike HSRP (must explicitly add preempt)
- Faster default timers: Hello=1s | Dead=3s — quicker failover than HSRP v1



POP QUIZ:

Which first-hop redundancy protocol is an open standard that works across multiple vendors?

- A. HSRP
- B. GLBP
- C. VRRP
- D. VIRDP



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DHCP — ENTERPRISE CONFIGURATION

DHCP CONCEPTS & IOS SERVER CONFIGURATION

- DHCP automates IP assignment — eliminates manual errors at scale; provides IP, mask, gateway, DNS, lease time
- DORA: Discover (client broadcast) → Offer (server) → Request (client) → Acknowledge (server confirms)
- IOS DHCP server: ALWAYS exclude static IPs FIRST: `ip dhcp excluded-address 192.168.10.1 192.168.10.20`
- Define pool: `ip dhcp pool VLAN10 → network 192.168.10.0 255.255.255.0 → default-router 192.168.10.1`
- Add DNS and lease: `dns-server 8.8.8.8 → lease 0 8 (8 hours)`
- Reservations (same MAC always gets same IP): `ip dhcp pool PRINTER → host 192.168.10.50 255.255.255.0 → hardware-address 0012.3456.7890`
- `show ip dhcp binding | show ip dhcp pool | show ip dhcp conflict`

DHCP RELAY & TROUBLESHOOTING

- Problem: DHCP Discover is a broadcast — routers block broadcasts by default
- DHCP Relay: configured on router interface facing clients — ip helper-address 10.0.0.10
- Relay converts broadcast to unicast directed at DHCP server; GIADDR field tells server which scope to use
- Multiple helper-address entries supported per interface — also forwards TFTP, NTP, and other UDP broadcasts
- Troubleshooting: client shows 169.254.x.x (APIPA) → check ip helper-address, routing to server, pool exhaustion
- debug ip dhcp server events — trace DORA process in real time (use sparingly in production)
- clear ip dhcp conflict * — flush conflict table when IPs in use weren't released properly

POP QUIZ:

What does ip helper-address do when configured on a router interface?

- A. Assigns DHCP addresses to clients on the interface
- B. Forwards DHCP broadcast messages to a specified unicast server IP
- C. Blocks DHCP traffic from unauthorized clients
- D. Configures the router as the DHCP server for the subnet



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POP QUIZ:

What is the correct order of the DHCP DORA process?

- A. Offer, Request, Acknowledge, Discover
- B. Discover, Offer, Request, Acknowledge
- C. Request, Offer, Discover, Acknowledge
- D. Discover, Request, Offer, Acknowledge



POP QUIZ:

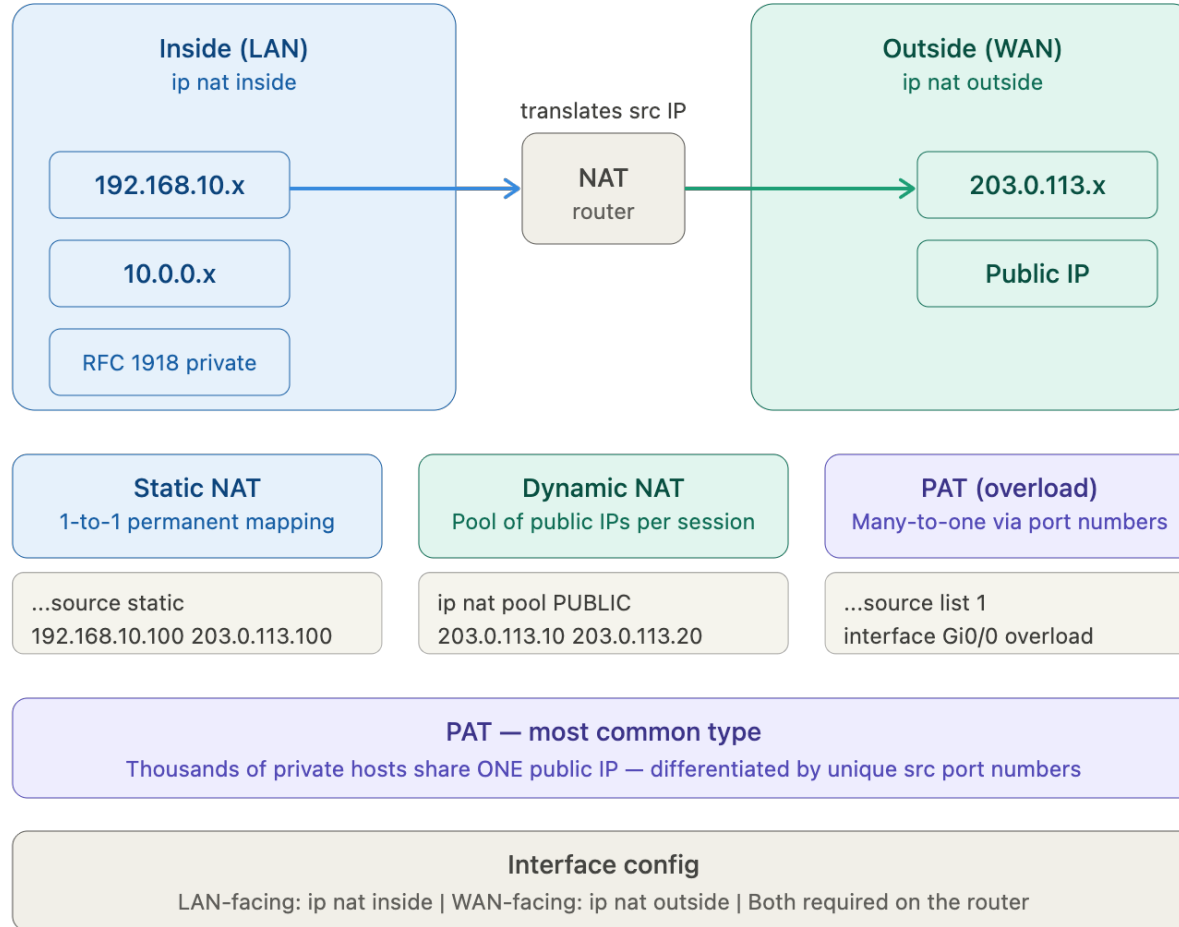
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- B. Discover, Offer, Request, Acknowledge
- C. Request, Offer, Discover, Acknowledge
- D. Discover, Request, Offer, Acknowledge



NAT TYPES & CONFIGURATION

NAT TYPES & CONFIGURATION



- NAT translates RFC1918 private IPs to public IPs at the network boundary
- Inside interface (LAN-facing): `ip nat inside` | Outside interface (WAN-facing): `ip nat outside`
- Static NAT: 1-to-1 permanent mapping — for servers needing consistent inbound public access
- Dynamic NAT: pool of public IPs assigned per-session
- PAT (NAT Overload): thousands of private IPs share ONE public IP using port numbers — most common type

NAT VERIFICATION & TROUBLESHOOTING

- show ip nat translations — active mappings: Inside Local → Inside Global → Outside Global → Outside Local
- show ip nat statistics — interface labels (inside/outside), total active translations, hits, misses
- debug ip nat — trace real-time translation events (use with caution in production environments)
- clear ip nat translation * — flush all dynamic translations for testing purposes
- Port forwarding (Static PAT): ip nat inside source static tcp 192.168.10.100 80 203.0.113.1 80
- Common issues: wrong nat inside/outside labels on interfaces, ACL not matching intended traffic
- Asymmetric routing breaks NAT — inbound and outbound traffic MUST traverse the same NAT device

POP QUIZ:

Which type of NAT allows thousands of internal devices to share a single public IP address?

- A. Static NAT — one-to-one permanent mapping
- B. Dynamic NAT — shared pool of public Ips
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- A. show ip nat pool
- B. show ip nat statistics
- C. show ip nat translations
- D. show ip interface brief



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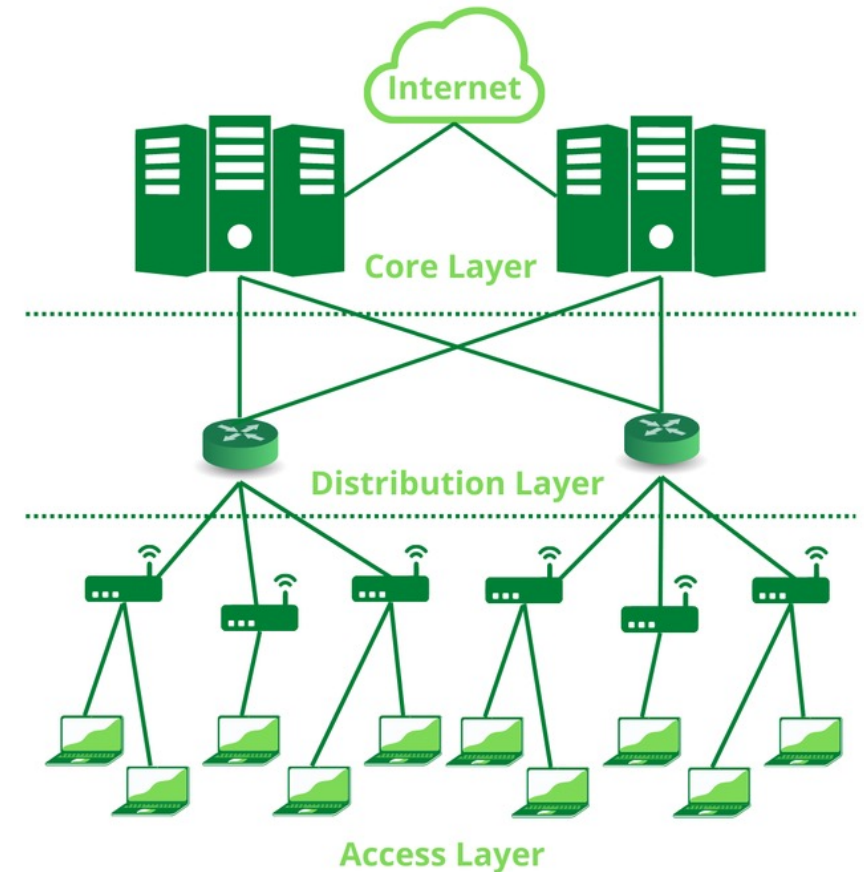
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DATA CENTER NETWORK ARCHITECTURE AND OPERATION

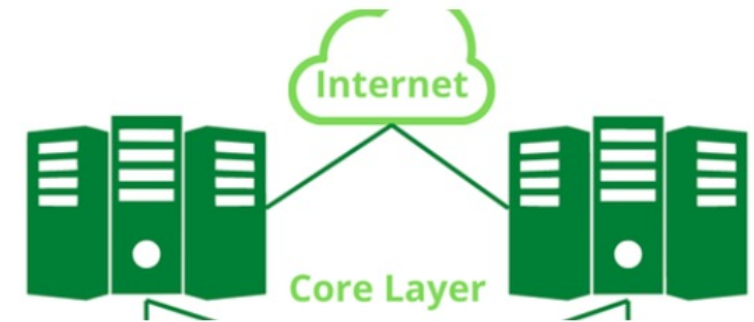
DATA CENTER NETWORK ARCHITECTURE OVERVIEW

- Hierarchical design provides scalability and performance
- Key layers:
 - Core Layer
 - Distribution Layer
 - Access / Edge Layer
 - Spine–Leaf Architecture (modern design)



CORE LAYER

- High-speed backbone of the network
- Provides fast, resilient connectivity between distribution switches
- Focus on throughput, redundancy, and minimal latency
- Typically no traffic filtering or packet manipulation



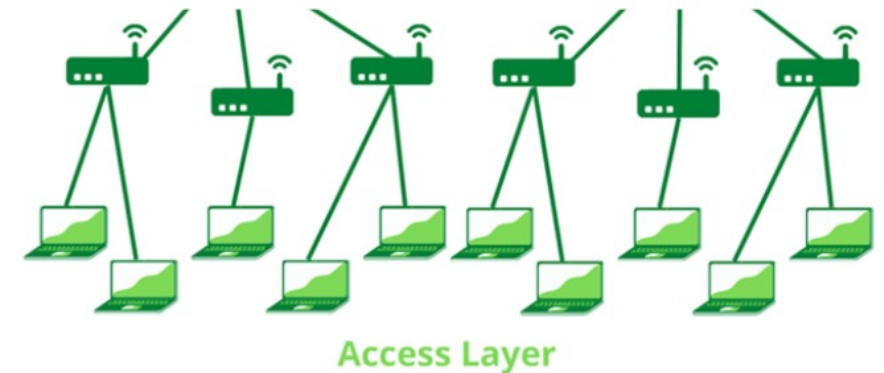
DISTRIBUTION LAYER

- Connects access (edge) layer to the core
- Enforces policy, security, and routing decisions
- Handles inter-VLAN routing, load balancing, and redundancy
- Often where first-hop redundancy protocols (HSRP/VRRP) operate

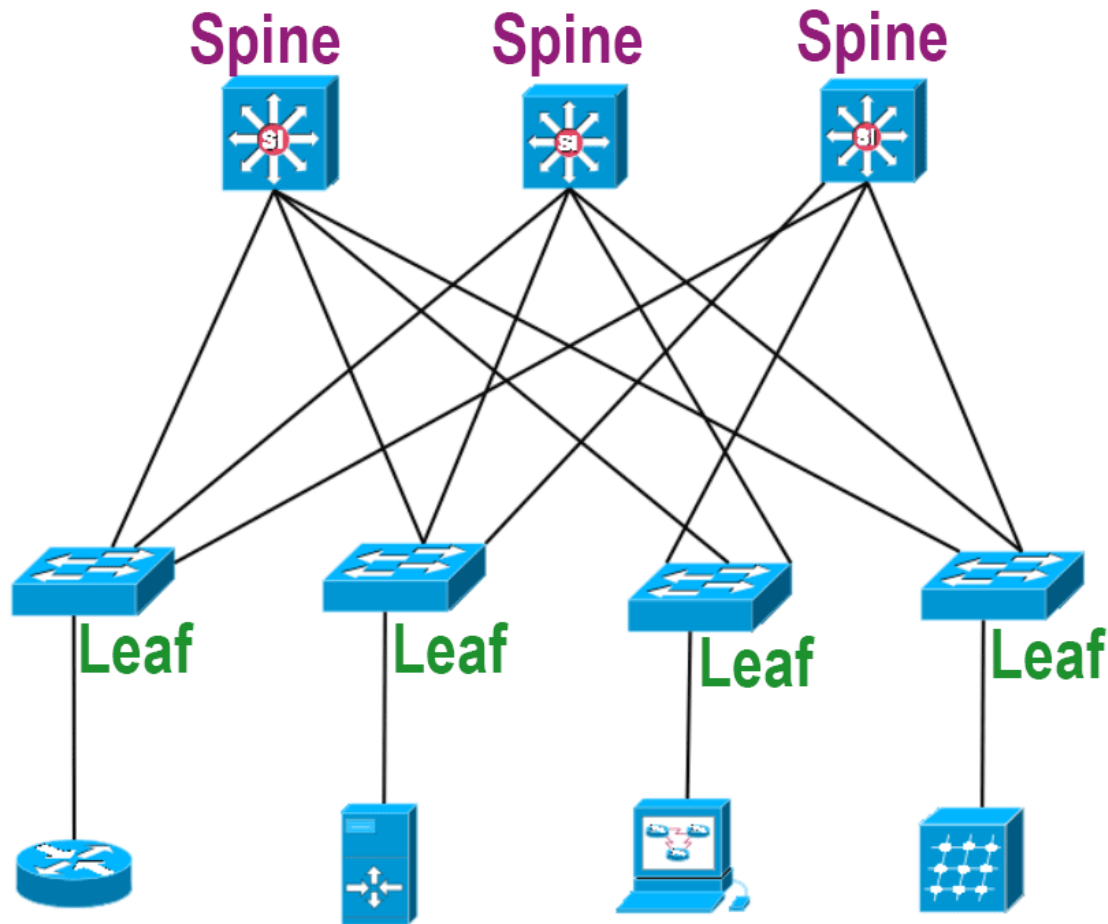


EDGE (ACCESS) LAYER

- Connects end devices, servers, and storage
- Provides port security, VLANs, QoS, and PoE
- L2 switching is common
- May use link aggregation for redundancy



SPINE-LEAF ARCHITECTURE



- Modern flat network design for data centers
- Spine switches: core backbone, high-speed, interconnect all leafs
- Leaf switches: connect to servers, storage, and other devices
- Every leaf connects to every spine — no single point of failure

DOCUMENTATION STANDARDS

- Accurate documentation = operational efficiency
- Includes:
 - Network diagrams (logical & physical)
 - IP address plans
 - VLAN and subnet lists
- Routing & device configurations
- Change logs and version control
- Use centralized repositories (e.g., Git, CMDB, or wiki)



CHANGE & INCIDENT MANAGEMENT

- Change Management:
 - Assesses, approves, and schedules network changes
 - Uses Change Advisory Board (CAB) review
 - Tracks impact, risk, and rollback plans
- Incident Management:
 - Detects and resolves unplanned disruptions
 - Uses ticketing, priority levels, and post-incident review (PIR)
 - Goal: Stability + Continuous Improvement

POP QUIZ:

What is the main role of the core layer in a hierarchical network?

- A. Enforce access policies
- B. Provide high-speed, resilient backbone connectivity
- C. Manage VLAN assignments
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POP QUIZ:

Which data center architecture eliminates single points of failure?

- A. Three-tier design
- B. Mesh topology
- C. Spine–Leaf architecture
- D. Star topology



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POP QUIZ:

At which layer are security and routing policies typically enforced?

- A. Core layer
- B. Access layer
- C. Distribution layer
- D. Spine layer



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POP QUIZ:

What is the main goal of Change Management in Network context?

- A. To troubleshoot network loops
- B. To approve and schedule controlled network modifications
- C. To monitor link aggregation status
- D. To configure routing protocols



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LAB 05: INTER-VLAN ROUTING WITH A MULTILAYER SWITCH (SVIS)

- Goal: Enable inter-VLAN routing using a multilayer switch (MLS) and Switch Virtual Interfaces (SVIs).
- Steps:
 - Create VLANs on the multilayer switch and access switches.
 - Assign PC access ports to the correct VLANs.
 - Configure trunk links between the access switches and the MLS.
 - Create SVIs for VLANs 10 and 20 on the MLS.
 - Assign IP addresses to the SVIs.
 - Enable Layer 3 routing using ip routing.
 - Configure PC IP addresses and default gateways.
 - Test inter-VLAN connectivity and verify routing.

LAB 06: FIRST-HOP REDUNDANCY WITH HSRP

- Goal: Provide a redundant default gateway using HSRP for hosts in VLAN 10.
- Steps:
 - Create VLAN 10 on SW1 and assign PC1's port to it.
 - Configure trunk links from SW1 to both routers.
 - Create router subinterfaces for VLAN 10 with unique IPs.
 - Configure HSRP group 10 with virtual IP 192.168.10.1.
 - Set router priorities and enable preemption (R1 active, R2 standby).
 - Configure PC1 to use the HSRP virtual IP as its default gateway.
 - Test connectivity and verify HSRP state.
 - Simulate failover by shutting down the active router's subinterface.
 - Restore the router and verify preemption behavior.

LAB 07: VLAN AND TRUNK TROUBLESHOOTING SCENARIO

- Goal: Identify and correct VLAN and trunk misconfigurations using systematic troubleshooting.
- Steps:
 - Document VLAN assignments and access port settings on both switches.
 - Review trunk status and allowed VLAN lists on Fa0/24.
 - Test connectivity between PCs to confirm the problem.
 - Identify incorrect or missing VLANs and port configurations.
 - Fix access VLAN assignments, create missing VLANs, and correct trunk allowed VLANs.
 - Re-verify VLANs, switchport modes, and trunk configuration.
 - Retest connectivity and confirm successful communication between PCs.
 - Write a troubleshooting report summarizing findings and fixes.